Pitch Length L Calculation Between The Helical spirals of the Prototype Magic Angle Solenoid

Clarification

1. **Pitch Length Definition**:
   - L is the distance between two adjacent helical turns *along the central axis* of the solenoid.
   - This is analogous to the "advancement" of a screw after one full rotation.

2. **Relationship to Pitch Angle**:

   The pitch angle (θpitch) is the angle between the **helical wire** and the **axis** of the solenoid. It satisfies:

   $$\tan(\theta_{\text{piteh}}) = \frac{\text{Circumference}}{\text{Pitch Length}} = \frac{2\pi R}{L}.$$

   For $\theta_{\text{pitch}} = \theta_m = \arctan(\sqrt{2}) \approx 54.74^\circ$, the Magic Angle θ_m , rearrange to solve for L :

   $$L = \frac{2\pi R}{\sqrt{2}} = \pi R\sqrt{2}.$$

   For R=0.51 cm radius of our prototype helical solenoid we get a pitch step $L \approx 2.27\ cm$.